TRAN THI THANH HOA

Research and Development Engineer – HMI and VR

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SUMMARY

PhD in **Human Factors and Human-Machine Interaction** with strong experience in operator activity analysis, usability evaluation, and interface design for complex systems. Skilled in combining **qualitative and quantitative methods** (interviews, observations, NASA-TLX, eye-tracking) to assess user needs, workload, and risks. Experienced in **simulation environments** (AR/VR), prototyping, and collaboration with multidisciplinary engineering teams. Strong ability to translate user insights into actionable design recommendations and system requirements. Fluent in English and French.

SKILLS

Human Factors and HMI Evaluation

- Activity analysis, operator interviews, usability testing, eye-tracking (Tobii), writing UI requirements and recommendations.
- Workload assessment (NASA-TLX), behavioral and cognitive analysis.
- Paper mock-ups, interface prototyping for passenger, design guidelines and usability improvement

Technical Skills

- Design & Prototyping: AR/VR prototyping (Unity, Davinci Resolve, Meta Quest Pro, HTC Vive, Valve), 3D modeling (SolidWorks, Catia, blender)
- Programming & Data Analysis: MATLAB, Python, R, C++, C#, GitHub
- Version Control & Continuous Integration: Git, GitHub

Methodology & Collaboration

- Agile methods & R&D-oriented workflows
- Mentoring junior developers and cross-functional teamwork
- Languages: French (fluent), English (fluent), Vietnamese (native speaker)

PROFESSIONAL EXPERIENCE

Researcher – Human-Machine Interaction
IRL Crossing CNRS, IMT Atlantique, Brest, France & Adelaide University, Australia | *09/2022 – 10/2025*

- Conducted **7 user studies** on trust, passenger experience and usability in autonomous vehicles using AR/VR interfaces.
- Performed workload and cognitive analysis using NASA-TLX, eye-tracking, and physiological data (GSR).
- Defined HMI requirements and design recommendations based on experimental results.
- Designed and developed a **VR autonomous vehicle simulator** for prototyping and testing novel AR interface visualizations.
- Published and presented results at **IEEE VR, ISMAR, ICAT**, contributing to global research on AR/UX.

- Tools: Unity, C#, Git, Python, R, Blender, Tobii Eye Tracker, Quest Pro, Davinci Resolve.

R&D Engineer – C++ / Virtual Reality
TechViz, Paris, France | *02/2022 – 07/2022*

- Collaborated with engineering teams to design and improve an automated testing platform.
- Collected and analyzed user feedback to enhance usability and workflow efficiency.
- Tools: C++, C#, JavaScript, Git, Redmine, Bash, HTC Vive, Valve, Catia.

Research Engineer – Graduation Project
CNRS, Grenoble, France | *02/2021 – 07/2021*

- Designed an **immersive VR modeling system** supporting direct 3D modeling operations (resize, translate, rotate, Boolean operations).
- Oversaw the entire development cycle: concept, prototype, user study, and publication
- Presented outcomes at the **CAD 2021 conference**.
- Tools: Unity, C#, Git, Python, SteamVR, FreeCAD, HTC Vive, Creo.

Team Leader – AVATAR Project (Erasmus+)
Grenoble, France | *02/2021 – 06/2021*

- Led a team of 5 students from 4 European universities on a **digital twin VR simulation project**.
- Managed project timeline, tasks, code reviews, and cross-cultural collaboration
- Contributing to an open-source 3D modeling program in VR, code reviews, and commits.
- Tools: Unity, Git, Catia, ROS, MATLAB.

EDUCATION

PhD in Computer Science (Human-Machine Interaction & UX)
IMT Atlantique, France & Adelaide University, Australia | *2022 – 2025*
France–Australia Cotutelle Program.

Engineer's Degree – Product Engineering
Grenoble INP, France | *2018 – 2021*
Awarded **Eiffel Excellence Scholarship** from the French Ministry of Europe and Foreign Affairs.

Engineer's Degree – Automated Production Engineering
Danang University of Technology, Vietnam | *2015 – 2018*
Top 1 of 80 students.

CERTIFICATIONS

- ERAMUS+ Certificate for Advanced Virtual and Augmented Reality Toolkit for Learning
- Coursera: 3D Interaction Design, 3D Models, Building Interactive 3D Characters and Social VR from the University of London in the field of virtual reality

  Machine Learning from the Standford University